

Exploratory Data Analysis Report

Customer Personality & Marketing Analytics Dataset

AI Lab 99 Internship Program 2026	
Module	ML-M04 — Exploratory Data Analysis
Department	Machine Learning
Dataset Size	2237 customers x 33 columns
Missing Values	0
Duplicate Records	0

1. Dataset Overview

The dataset contains **2237 customer records** across **33 columns**, covering demographics, purchase history, marketing campaign responses, and website engagement.

Missing values: 0 (none) **Duplicate records:** 0 (none). The dataset supplied was pre-cleaned, requiring no further imputation or deduplication before analysis.

Key Fields

ID, Year_Birth/Age, Education, Marital_Status, Income, Kidhome, Teenhome, Dt_Customer, Recency, spending fields (Wines, Fruits, Meat, Fish, Sweets, Gold), purchase-channel counts (Web/Catalog/Store/Deals), NumWebVisitsMonth, campaign acceptance flags (AcceptedCmp1-5, Response), and Complain.

2. Customer Demographic Analysis

Average customer age is **57.1 years** (median 56.0). Average income is **\$51,855** (median \$51,382). The most common education level is **Graduation**, and the most common marital status is **Married**. **71.5%** of customers have at least one child or teenager at home.

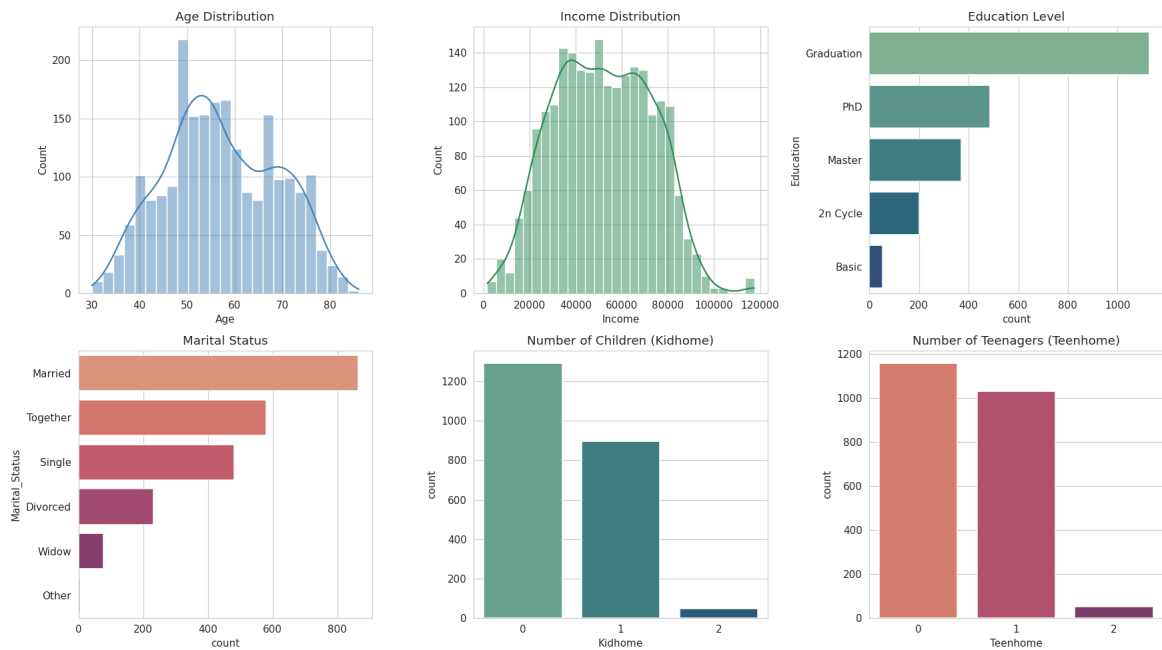


Figure 2.1 — Age, income, education, marital status, and household composition

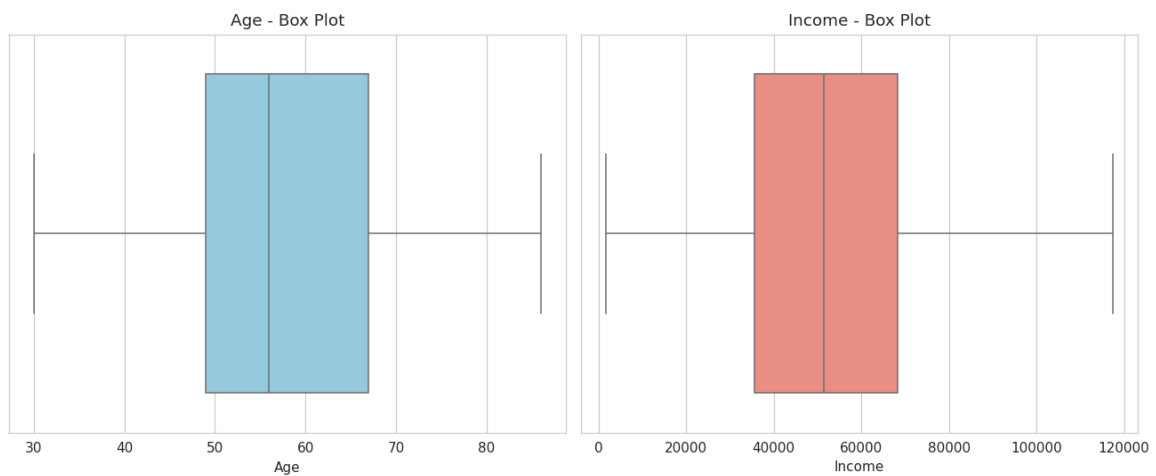


Figure 2.2 — Age and income box plots (outlier check)

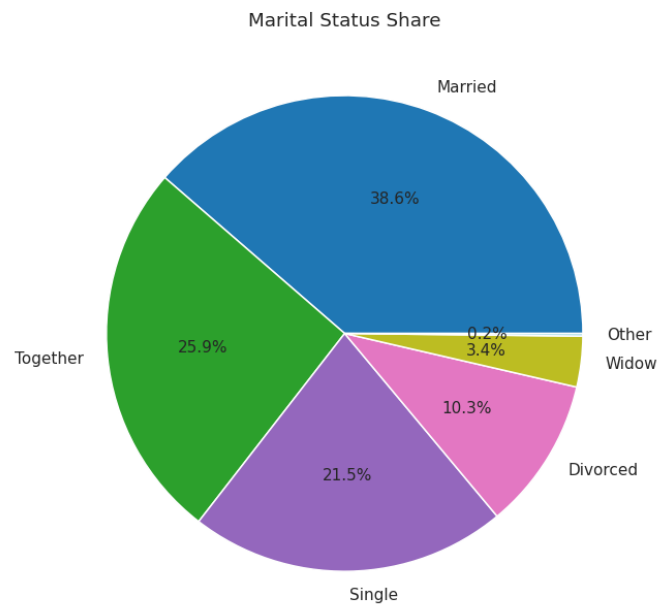


Figure 2.3 — Marital status share

3. Customer Spending Analysis

Total spending across all customers is **\$1,270,974**, averaging **\$568** per customer.

Spending by Product Category

Category	Total Spend
Wines	\$675,920
MeatProducts	\$337,387
GoldProds	\$88,299
FishProducts	\$71,821
SweetProducts	\$49,073
Fruits	\$48,474

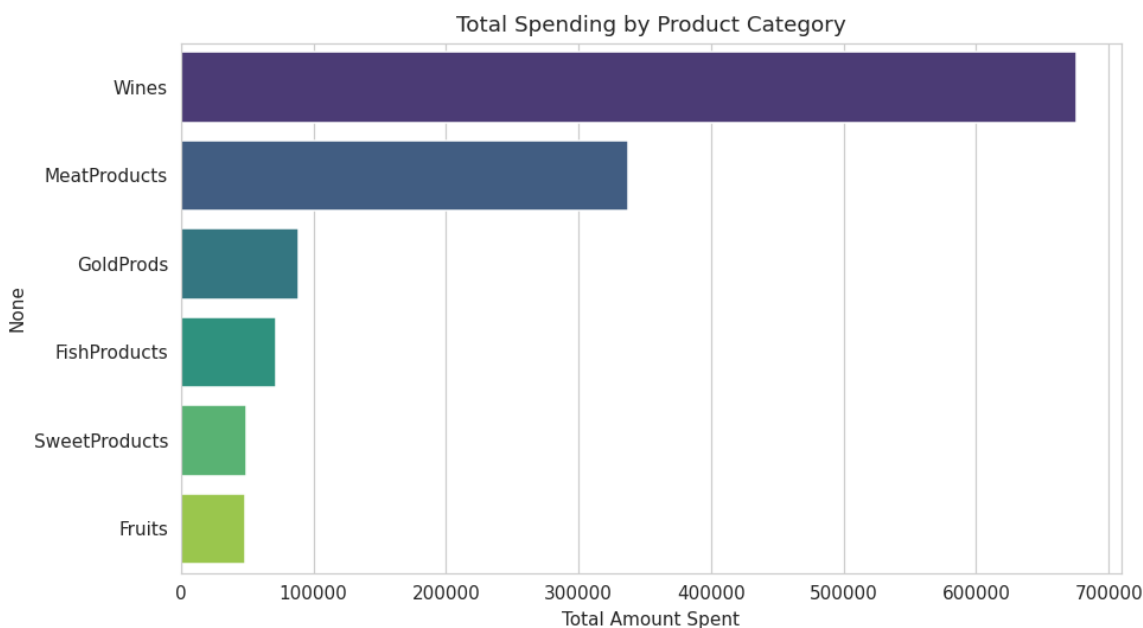


Figure 3.1 — Total spending by product category

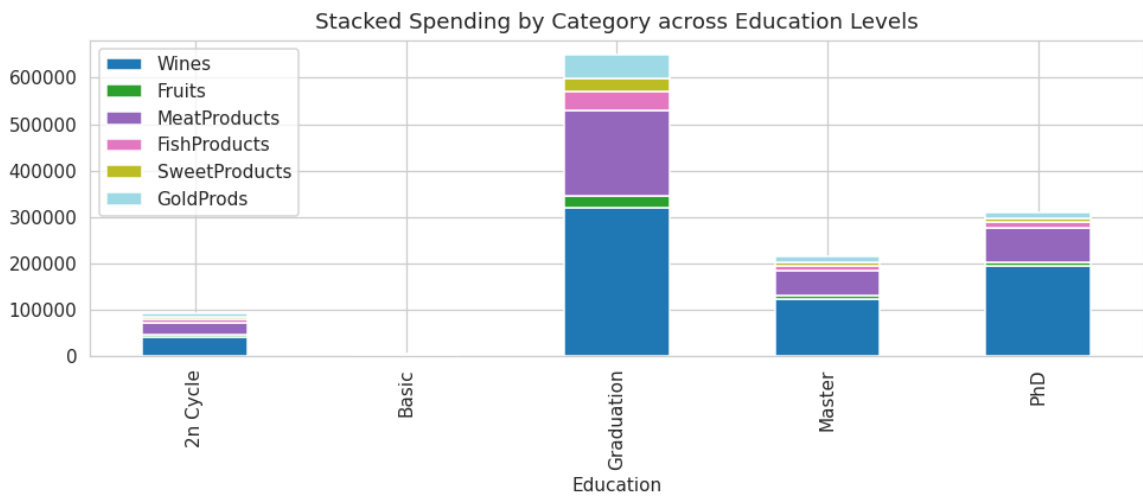


Figure 3.2 — Stacked spending by category across education levels

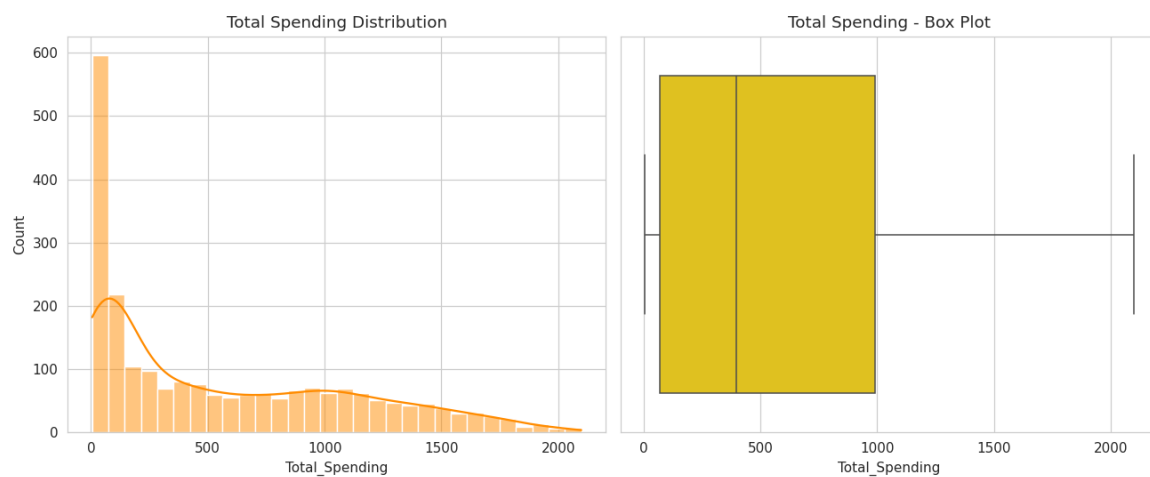


Figure 3.3 — Total spending distribution and box plot

4. Purchasing Behavior Analysis

Store purchases lead all channels with **12,962** total transactions, followed by web (9,143), catalog (5,956), and deal-based purchases (5,205).

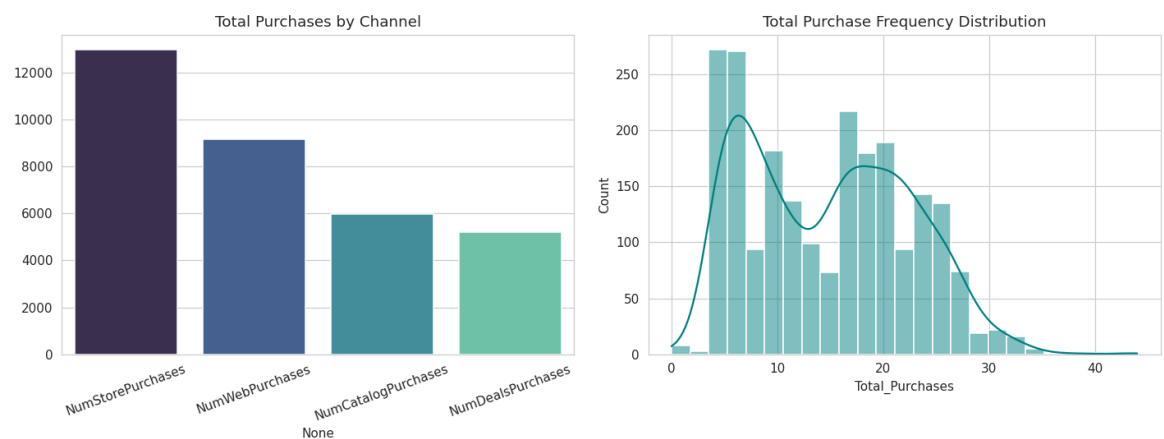


Figure 4.1 — Purchases by channel and purchase-frequency distribution

5. Website Engagement Analysis

Customers visit the website an average of **5.32** times per month. **19.5%** are frequent visitors (8+ visits/month), while **16.3%** show low engagement (2 or fewer visits/month).

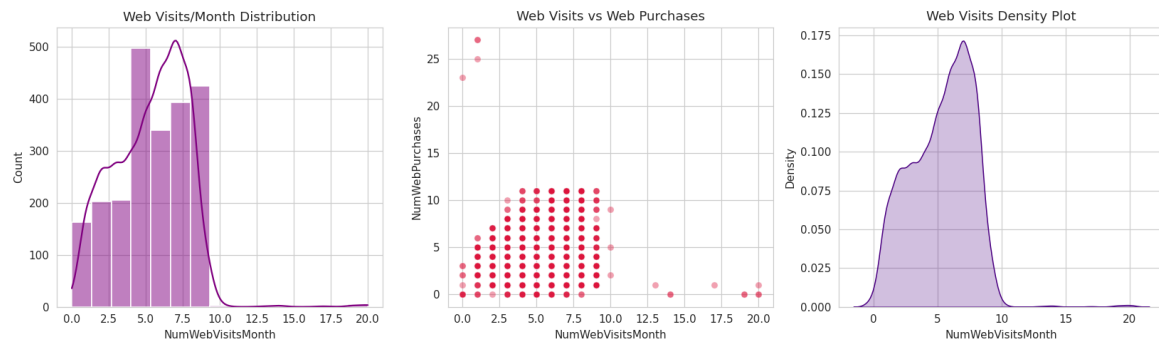


Figure 5.1 — Web visit distribution, visits vs purchases, and density plot

Frequent website visits do not strongly translate into more web purchases, suggesting browsing behavior and buying behavior are only loosely linked.

6. Customer Recency Analysis

Average recency (days since last purchase) is **49.1** days. **32.3%** of customers are active (purchased within 30 days), while **38.5%** are inactive (60+ days since last purchase).

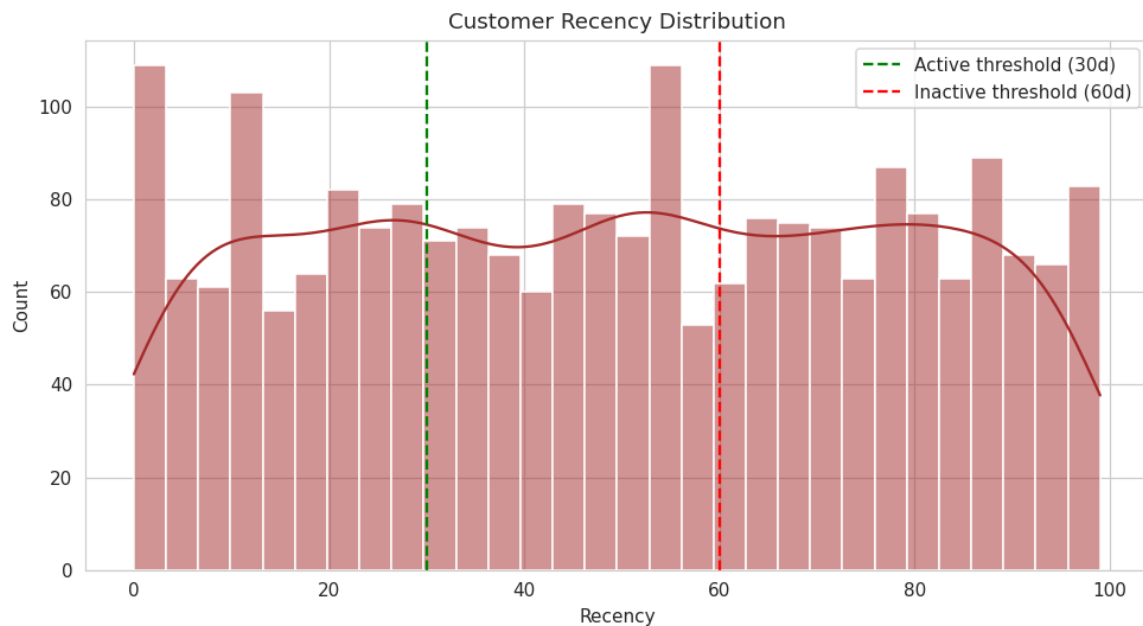


Figure 6.1 — Recency distribution with active/inactive thresholds

Business Insight: A substantial inactive segment exists and should be prioritized for re-engagement campaigns (e.g., win-back offers, personalized reminders).

7. Marketing Campaign Analysis

Acceptance Rates

Campaign	Accepted	Acceptance Rate
Campaign1	144	6.44%
Campaign2	30	1.34%
Campaign3	163	7.29%
Campaign4	167	7.47%
Campaign5	162	7.24%
Latest Campaign	334	14.93%

The best-performing campaign is the **Latest Campaign**, with the highest acceptance rate.

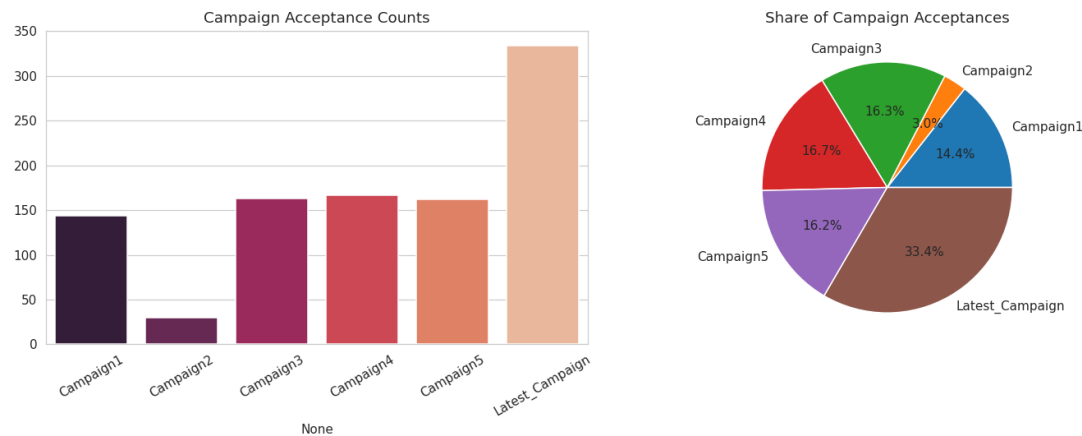


Figure 7.1 — Campaign acceptance counts and share

8. Customer Complaint Analysis

Complaints are rare: **20** customers (0.89%) lodged a complaint. Complainers show slightly lower average spending (\$383 vs \$570) and income (\$45,672 vs \$51,911) than non-complainers.

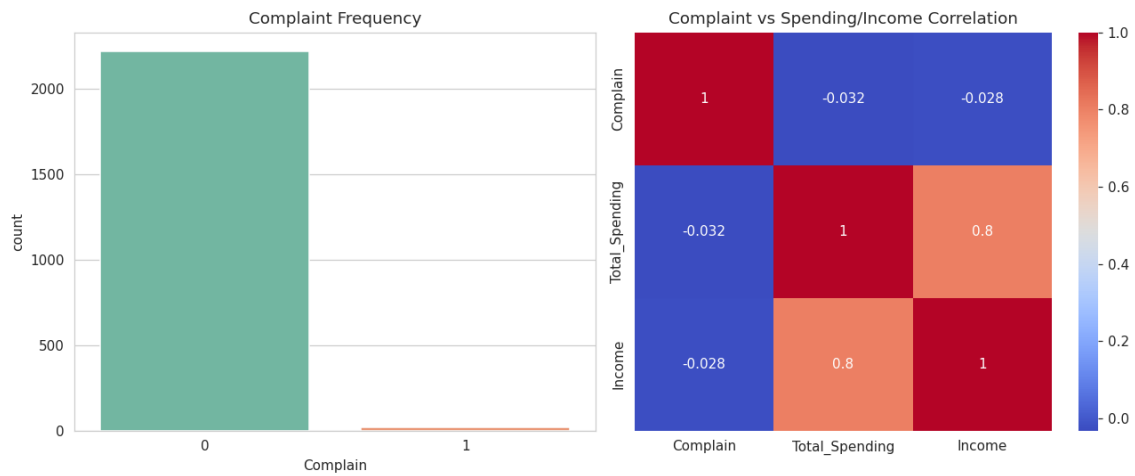


Figure 8.1 — Complaint frequency and correlation heatmap

9. Correlation Analysis

Strongest Positive Correlations

- Income vs Total_Spending: **0.803**
- Total_Spending vs Income: **0.803**
- Total_Spending vs Total_Purchases: **0.769**

Strongest Negative Correlations

- Total_Spending vs Kidhome: **-0.563**
- Income vs NumWebVisitsMonth: **-0.648**
- NumWebVisitsMonth vs Income: **-0.648**

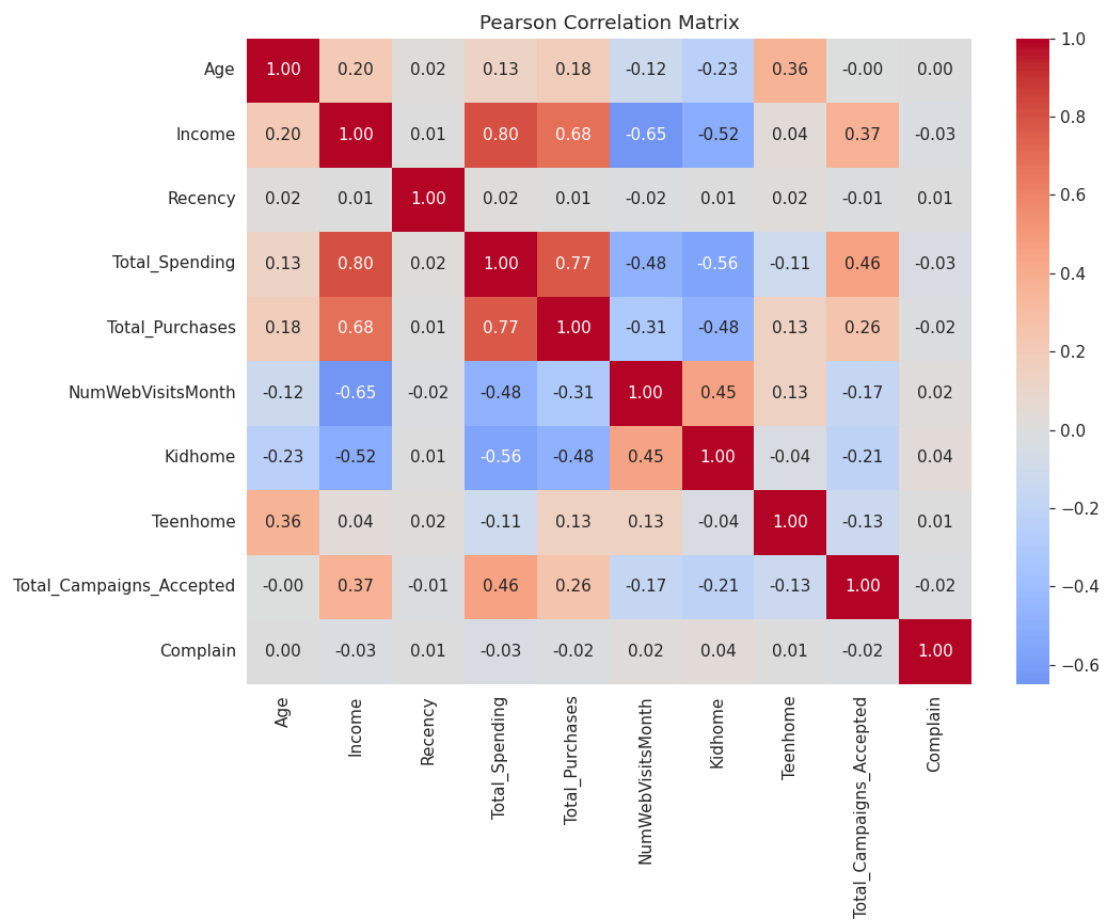


Figure 9.1 — Pearson correlation heatmap



Figure 9.2 — Income vs Total Spending scatter plot

10. Customer Segmentation

Customers were segmented by total spending quartiles: **561** High-Value, **1109** Standard, and **567** Low-Value customers.

Additional behavioral segments: **727** discount seekers, **607** frequent buyers, **462** campaign responders, **723** loyal/recent customers, and **861** inactive customers.

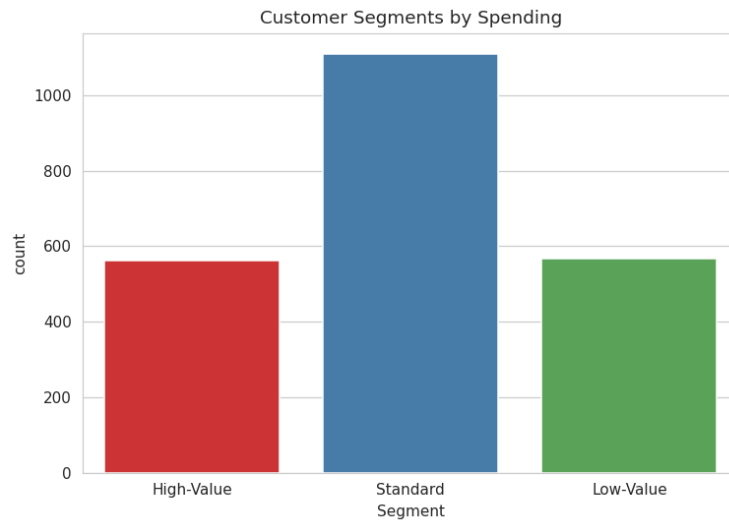


Figure 10.1 — Customer segments by spending

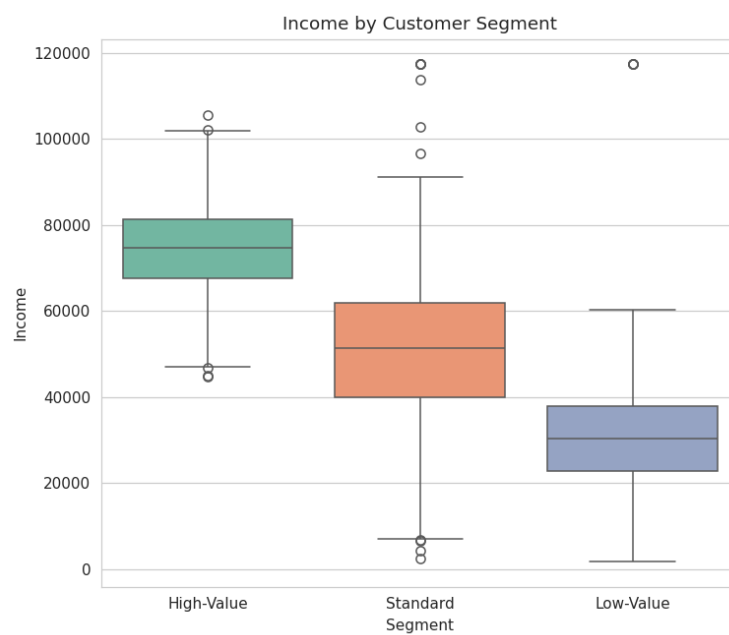


Figure 10.2 — Income distribution across segments

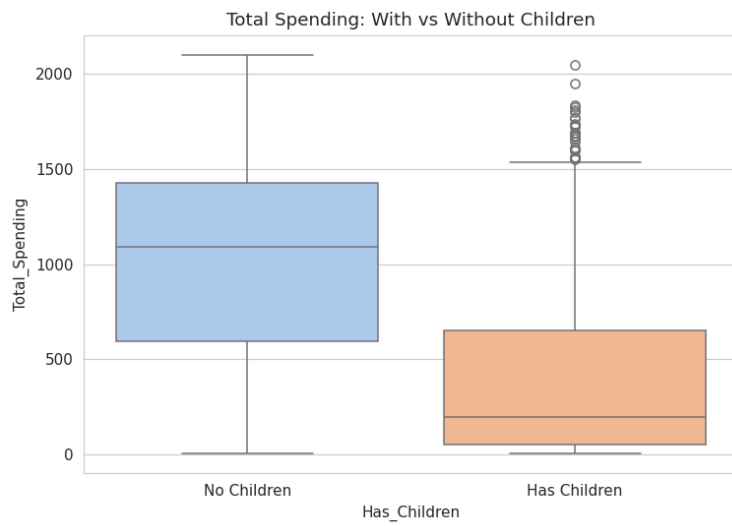


Figure 10.3 — Spending: customers with vs without children

Customers without children spend on average **\$1,003**, compared to **\$395** for those with children — roughly 2.5x more.

11. Business Insights Summary

Which customers spend the most?

High-Value segment: older, higher-income customers, mostly without children.

Which product category generates the highest revenue?

Wine, followed by Meat products.

Does income influence spending?

Yes — strong positive Pearson correlation (~0.80).

Which purchase channel is most popular?

Store purchases (in-store)

Which campaign performs best?

Latest Campaign — highest acceptance rate.

Which customers respond most frequently?

Existing high-spending, recently-active (loyal) customers.

Which customers are inactive?

~38.5% of customers (Recency > 60 days) — target for re-engagement.

Which customers visit the website most?

Lower-income customers tend to visit more often, without a proportional purchase lift.

Do customers with children spend differently?

Yes — they spend markedly less than customers without children.

What characteristics define high-value customers?

Avg age 58.5, avg income \$74,349, education mostly Graduation, prefers Store purchases, low complaint rate.

Which variables have the strongest correlations?

Income-Spending (+0.80), Spending-Purchases (+0.77), Income-WebVisits (-0.65), Kidhome-Spending (-0.56).

Recommended Marketing Strategies

- Launch targeted re-engagement campaigns for customers inactive 60+ days.
- Prioritize in-store and catalog promotions for high-value, older, higher-income segments.
- Bundle Wine and Meat products — the top revenue drivers — in cross-sell promotions.
- Design family-oriented offers/discounts to lift spending among customers with children.
- Replicate the design and targeting approach of the Latest Campaign, which achieved the highest acceptance rate.